

TOPIC 34/60

Vocabulary in Tokenizers

The Dictionary of the AI.

If a word, concept, or character is not in the vocabulary, it doesn't exist to the Language Model. Let's look under the hood at the master lookup table.

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THE MASTER LIST

What is the Vocabulary?

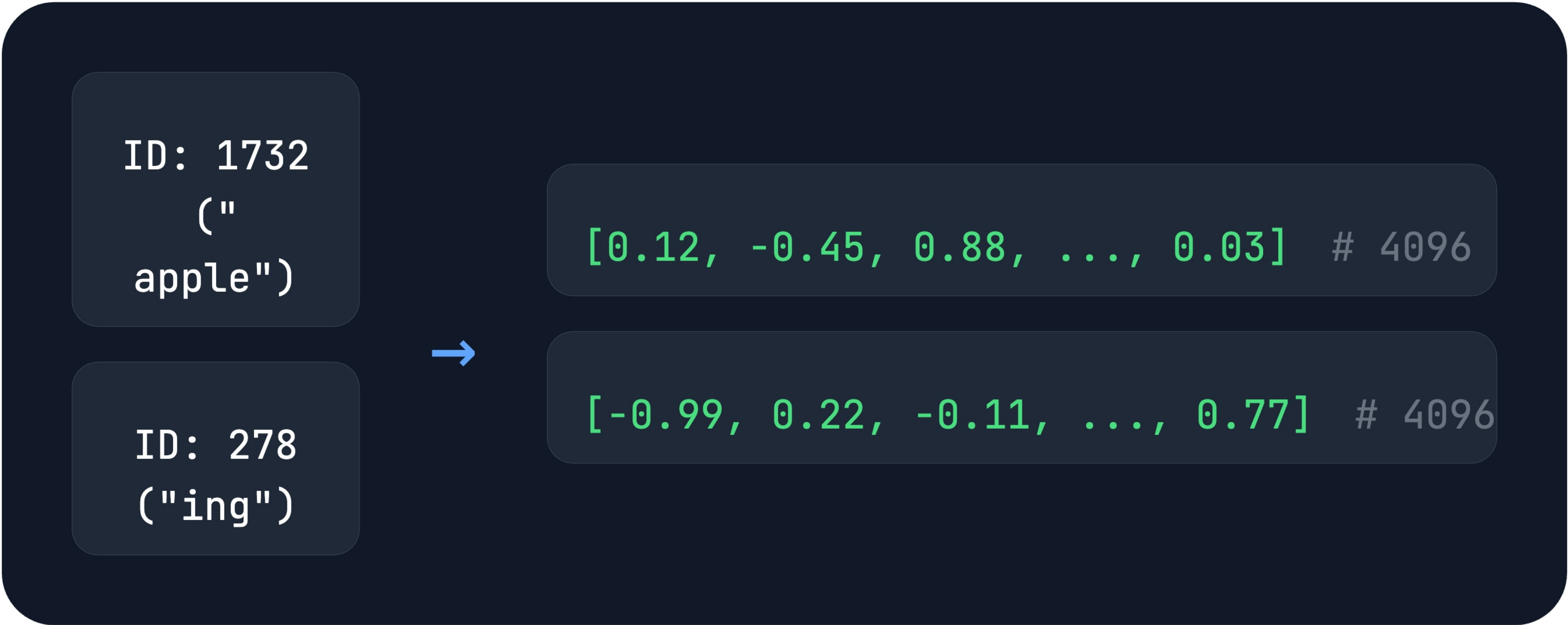
A neural network cannot process raw text. It processes numbers. The **Vocabulary** is a fixed, finite dictionary generated during the tokenization process (e.g., via BPE). It maps distinct text chunks to unique integer IDs.

Token String	Token ID
"a"	64
" apple"	1732
"ing"	278
" artificial"	14920



The Embedding Table

The vocabulary size dictates the shape of the model's very first layer: the **Embedding Matrix**. Every single Token ID must map to a high-dimensional mathematical vector (e.g., 4096 dimensions).



Formula: $\text{Memory Cost} = \text{Vocabulary Size} \times \text{Vector Dimension} \times \text{Parameter Precision}$. If your vocabulary is massive, your embedding table eats up gigabytes of VRAM.



THE ENGINEERING DILEMMA

How Big Should It Be?

Why not just include every word in the dictionary? Because vocabulary size is a direct trade-off between model efficiency and memory footprint.

Small Vocab (e.g. 30k)

- ✓ Tiny memory footprint
- ✓ Easier for the model to train
- ✗ Text gets shattered into many subwords
- ✗ Eats up the context window quickly
- ✗ Slower inference (more tokens to generate)

Large Vocab (e.g. 128k+)

- ✓ Highly efficient tokenization (few tokens per word)
- ✓ Faster inference speed
- ✓ Fits more text in context window
- ✗ Massive RAM requirement for Embedding Table
- ✗ Harder to train rare tokens

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EDGE CASES

Out of Vocabulary (OOV)

What happens if you feed the model a word it has never seen before, and it isn't in the vocabulary? It defaults to shattering the word into known subwords, or even single characters/bytes.

Input Word:

"Pseudohypoparathyroidism"

Tokenized Output (Shattered):

Pseudo

hypo

para

thyroid

ism

Older systems used a literal [UNK] (Unknown) token, which destroyed meaning. Modern byte-level BPE never uses [UNK]—it just falls back to individual byte tokens as a last resort.

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GLOBAL INEFFICIENCY

The Multilingual Penalty

If a model's vocabulary is built primarily on English data, it will contain full English words. For other languages, it has to piece words together character by character. **This makes using LLMs in non-English languages significantly more expensive.**

"Hello, how are you?"

English (6 Tokens)

Hello , how are you ?

Hindi: "नमस्ते, आप कैसे हैं?" (18 Tokens)

Because the Devanagari script isn't well-represented in the vocabulary, the tokenizer shatters the Hindi sentence into microscopic byte chunks. You pay 3x the API cost for the same meaning.

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STRUCTURAL CONTROL

Special Tokens

The vocabulary isn't just made of text. It contains hidden, structural control tokens that tell the LLM when to start, stop, or change roles. These are never seen by the user.

Common Special Tokens

`<|endoftext|>` (Stops generation)

`<|im_start|>` (Starts a message)

`[MASK]` (Used in BERT for blanks)

Under the Hood

When you type a prompt in ChatGPT, the system silently wraps your text in these vocabulary tokens before sending it to the neural network.

```
<|im_start|>user  
Hello!  
<|im_end|>
```

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ANOMALIES

The Glitch Token Phenomenon

Sometimes, tokenizers are trained on raw, unfiltered web datasets (like Reddit scrapings) that contain weird, repeating strings. However, the *neural network* isn't trained on that data.

"SolidGoldMagikarp"

This was a username of a frequent poster on Reddit. The BPE algorithm saw it so often, it merged it into a single Token ID in the vocabulary.

The Brain Break

When early GPT models were asked to repeat "SolidGoldMagikarp", they would output completely random words like "distribute".

Why? The token existed in the vocabulary, but its vector in the Embedding Table was completely untrained and random.

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STATE OF THE ART

The Push for Larger Vocabularies

MODEL FAMILY	VOCAB SIZE	IMPACT
GPT-2 (2019)	50,257	The baseline. Mostly English-centric.
GPT-4 (cl100k_base)	100,277	Doubled in size. Massively improved coding and multilingual support.
Llama 3 (2024)	128,256	Massive efficiency gains. Fits 15% more text in the same context window vs Llama 2.
Gemma / Qwen	150,000+	Pioneering hyper-multilingual and structural efficiencies.



Master the Foundation

WORDS DON'T EXIST. ONLY IDS.

The Vocabulary is the ultimate bottleneck. It defines how fast the model reads, how much memory it requires, and which languages it treats as native versus foreign.

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